- How was Quiz 1?
- HW2 posted.

Lecture 4.

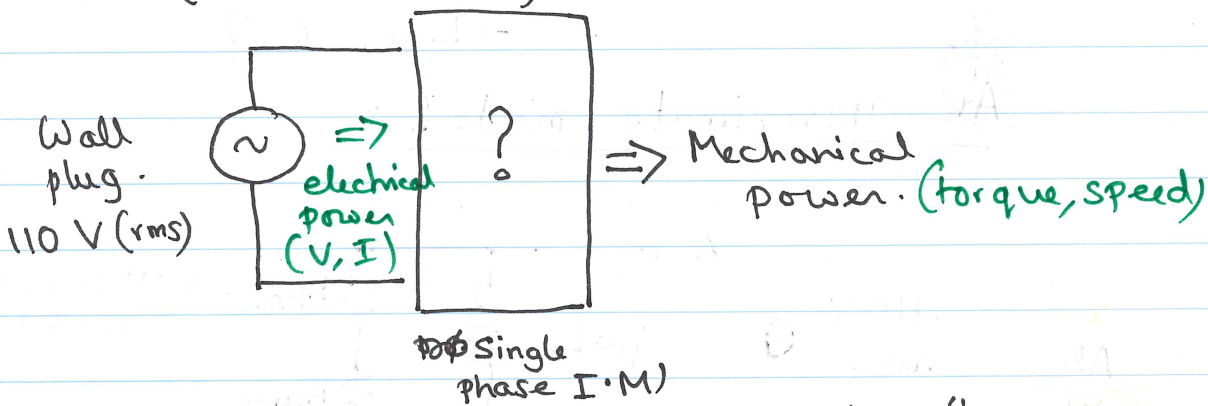
Suggested Reading – 2.5 - 2.7 (Pai).

Last time: * Conservation of power.
* Single phase circuits
* Power factor correction (Prof. Zhang).

Today: * Example of power factor correction.
* Concepts of three-phase power circuits.

A. Why do we need power factor correction?

→ Let us try to model a single-phase induction motor (circuit model). [IM].



Let us look at the "name-plate".

$V_1 = 110$ V, rms $I_1 = 11.6$ A, rms.

Output power: 1 HP (746 W) (mechanical power)

- We are interested in an approximate electrical circuit model.
- There is some conversion efficiency from electrical to mechanical power: 80% (assume).

∴ Electrical power input : $\frac{746}{0.8} = 932\ \Omega$.

Basically, you need to 932 Ω of electrical power to convert 746 Ω of mechanical power.

- How do we model the motor ? R, L, C ?
- What is the apparent power ? $S = V_{rms} I_{rms}$
  $= 1276\ VA$.

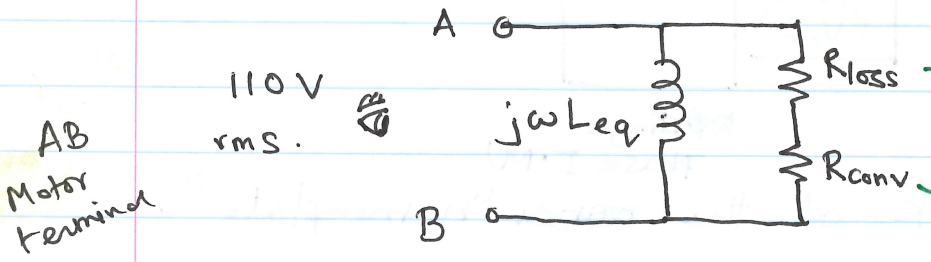
$S = 1276\ VA$

$P_{elec} = 932\ \Omega$

$P_{mech} = 746\ \Omega$.

As, $S \neq P_{elec}$. ⇒ Some reactive power ?
- L or C ?

<u>An approximate model :</u>



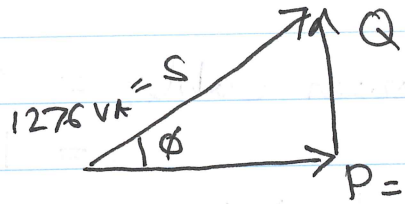
$R_{conv} = 10.4\ \Omega$

$R_{loss} = 2.6\ \Omega$

$R_{eq} = R_{loss} + R_{conv}$.

$= \frac{V^2}{P} = \frac{110^2}{932} = 13\ \Omega$.

$L_{eq} = ?$ $\quad X_{eq} = \omega L_{eq}$.



$Q = \sqrt{1276^2 - 932^2}$

$= 871$ VAR.

$\cos\phi = 0.75$ (lag).

$\therefore \quad X_{eq} = \frac{V^2}{Q} = 14\ \Omega$

$\therefore \quad L_{eq} = \frac{14}{2\pi \times 60} = 37$ mH

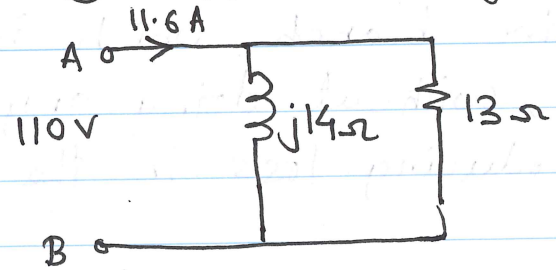
Detour: Proof:

$$\bar{S} = \bar{V}\bar{I}^* = \bar{V}\left[\frac{\bar{V}}{R} + \frac{\bar{V}}{jX}\right]^*$$

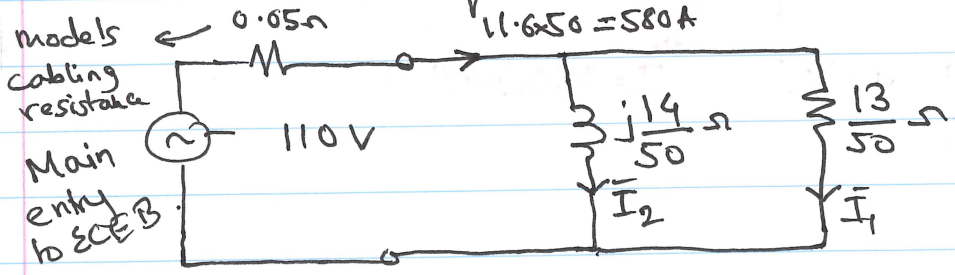
$$= \frac{\bar{V}^2}{R} + j\frac{\bar{V}^2}{X}$$

$$= P + jQ.$$

So, we have modeled our first IM.



* Now, let us imagine ECEB has 50 such motors. Net equivalent circuit:



Power Loss in power cables = 0.05×580^2 W

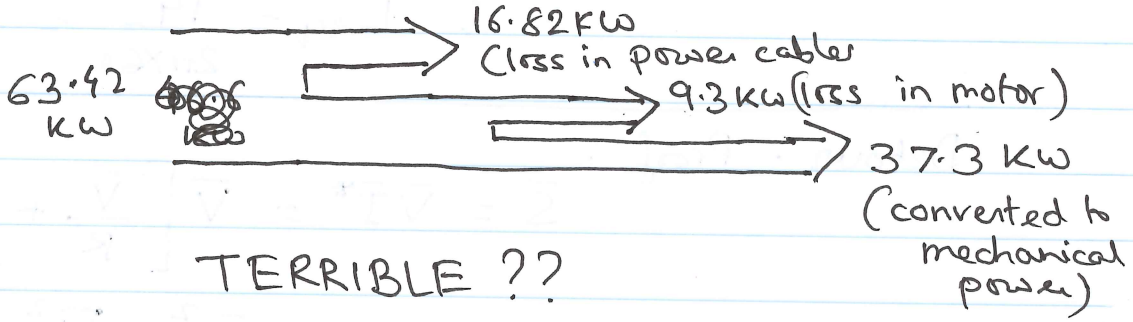
= 16.82 KW.

Useful mechanical power = 50×746 W

= 37.3 KW.

Drawn electrical power = 50×932 W

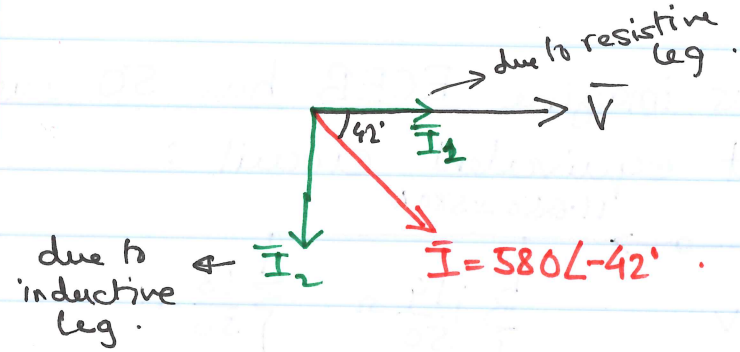
= 46.6 KW



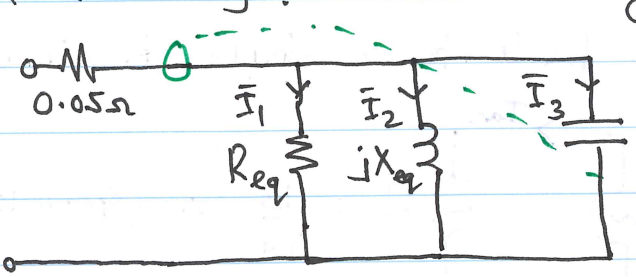
TERRIBLE ??

Q Can we improve, particularly, can we reduce loss in power cables ?

- Think about it this way, there is an inductive branch, which is drawing current ($\bar{I}_2$), without doing any useful work. but producing loss in the cable.



What happens if we could supply reactive power locally?



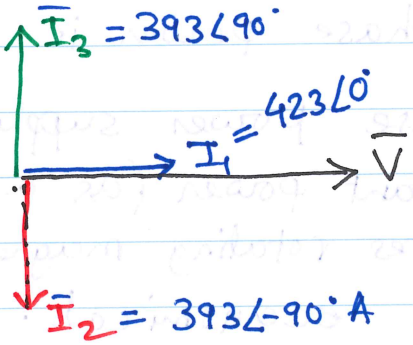
Objective: to minimize current drawn from the grid.

$$Q_{compensation} = 50 \times 871 \text{ VAR}$$
$$= 43.55 \text{ KVAR}.$$

$$X_c = \frac{110^2}{43.55 \times 10^3} = 0.28\ \Omega.$$

$X_c = \frac{1}{\omega C}$

$$\therefore \quad C_{cap} = \frac{1}{2\pi \times 60 \times 0.28} = 9.5 \text{ mF}.$$



$I_{tot} = 423\angle 0^\circ$

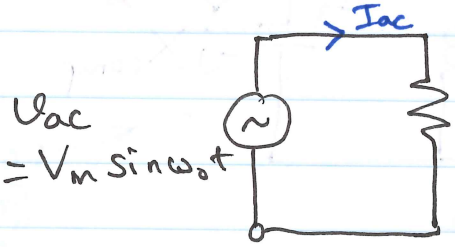
upf

lower than 580A.

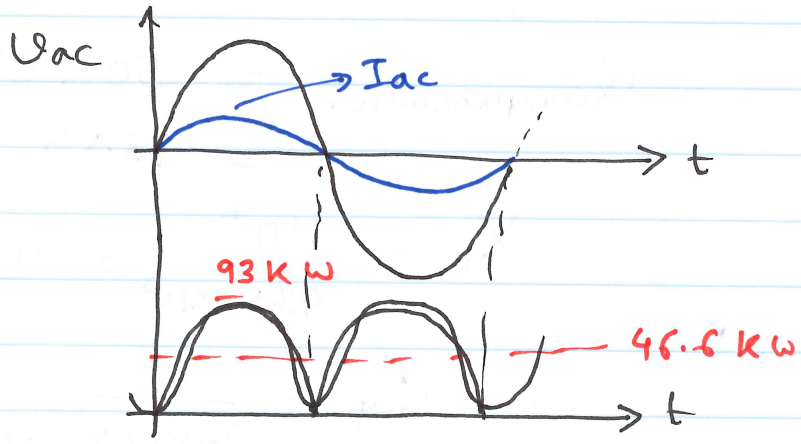
$\therefore$ loss in power cable
$= 423^2 \times 0.05$ W
$= 8.9$ KW
(cut the loss by 50%).

Concepts in three-phase power :

Let us go back to a simple R circuit.



How does this power look like ?



- Single phase power is oscillatory in nature?
- Three phase power supply enables.
  - constant power (as a function of time)
  - creates rotating magnetic field.
  - more economical.
    - less Cu (conductor material) to transmit same elec. power than equ. single phase. or two phase.

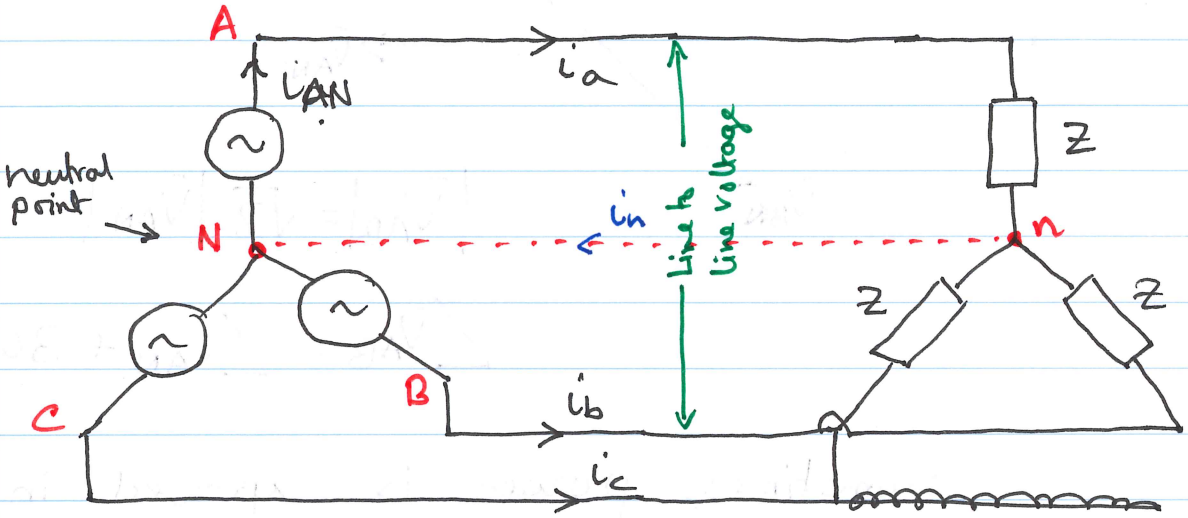
Current flowing in an arm = phase current

Voltage across an arm = phase voltage.

Three phase: Consider three ac sources which are phase shifted by 120°.

Phase voltage

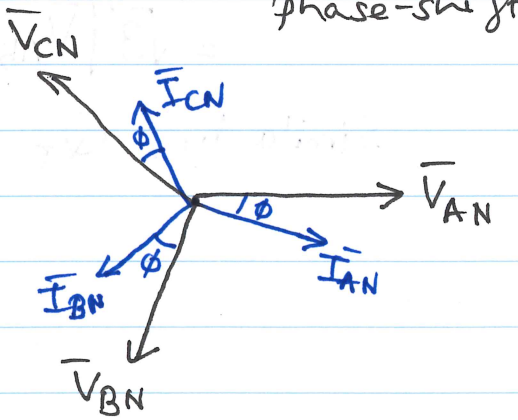
$$\left.\begin{aligned} v_{AN}(t) &= V_m \sin(\omega_0 t) \\ v_{BN}(t) &= V_m \cos(\omega_0 t - 120^\circ) \\ v_{CN}(t) &= V_m \cos(\omega_0 t + 120^\circ) \end{aligned}\right\}$$ Balanced source.



Three phase Y- source

Three-phase Y- balanced load.

As if three-single phase circuits phase-shifted.



$i_n = 0$

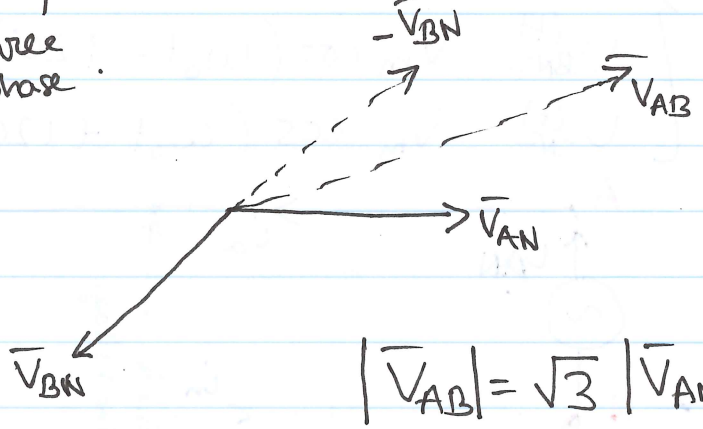
[if loads are balanced] all Z.

Power from each phase $\bar{V}_{AN}\bar{I}_A$

Total complex power:

$$\bar{S}_T = 3\,\bar{V}_{AN}\bar{I}_A$$

↑ three phase.



$$\bar{V}_{AB} = \bar{V}_{AN} - \bar{V}_{BN}$$

$$|\bar{V}_{AB}| = \sqrt{3}\,|\bar{V}_{AN}|$$

$$\angle\bar{V}_{AB} = \angle\bar{V}_{AN} + 30^\circ$$

Sometimes, power is expressed in line quantities.

$$P_T = 3\,|\bar{V}_{AN}|\,|\bar{I}_A|\cos\phi$$

($|\bar{V}_{AN}|$: phase voltage; ϕ: angle between phase voltage and phase current)

$$= \sqrt{3}\,|\bar{V}_{AB}|\,|\bar{I}_A|\cos\phi$$

($|\bar{V}_{AB}|$: line voltage)

Similarly, Q_T.